

The menopause

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Handwritten text in a cursive script, likely a reproduction of Leonardo da Vinci's notes on anatomy and proportions, is visible above the drawing of the Vitruvian Man.

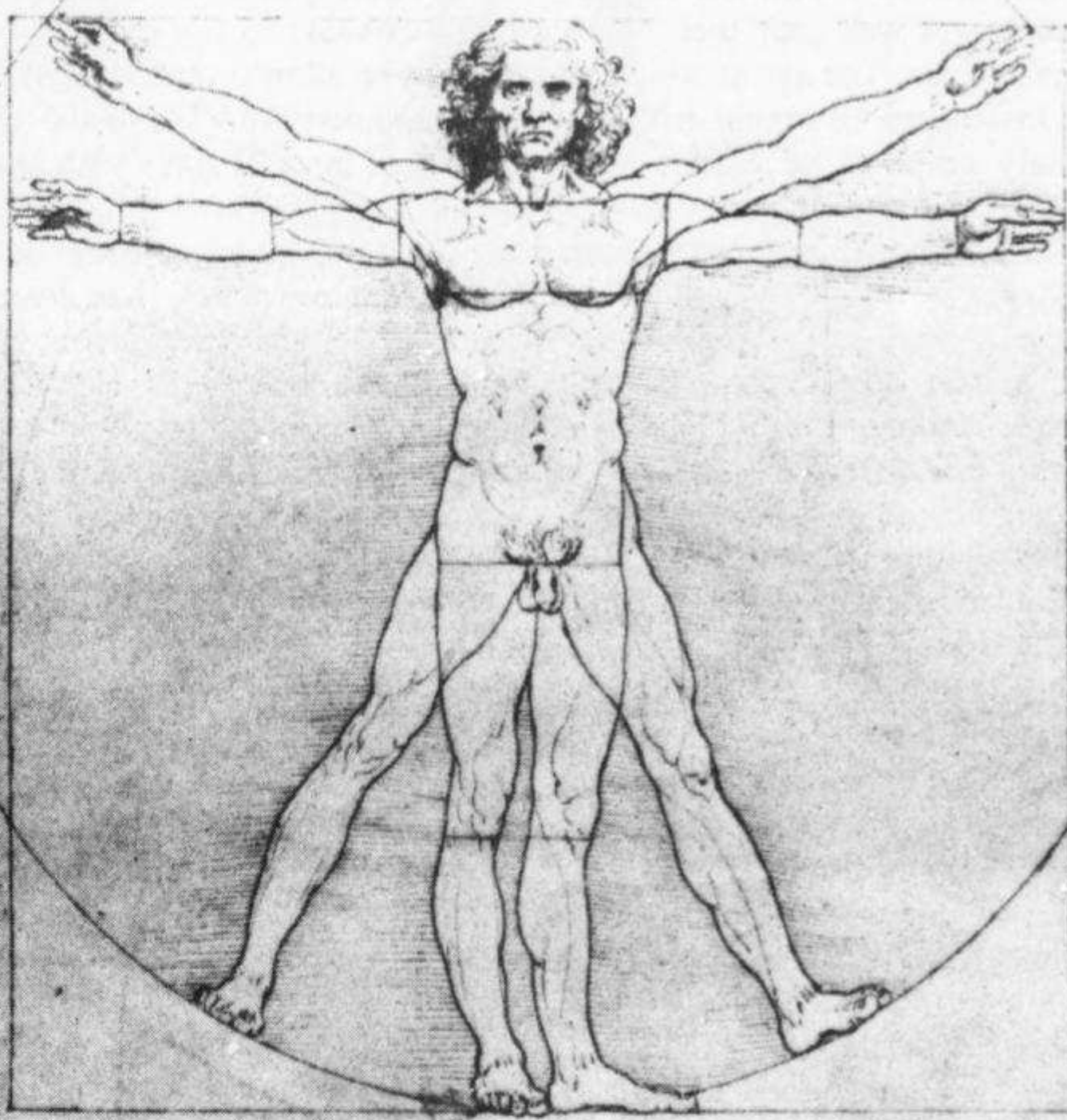


Figure 1: Although normal proportions, as shown in Leonardo da Vinci's "Universal Man", are for the height to equal the armspan, after the menopause there may be a decrease in height with no corresponding decrease in armspan, which may indicate oestrogen replacement.

WITH THE RECENT popularity of articles on the menopause in the Sunday heavies and glossy monthlies one may be forgiven for falsely imagining that all symptoms between 35 and 55 in either sex can be attributed to this single cause. The menopause is a natural stage in a woman's life — it is puberty in reverse — and it marks the end of her reproductive cycle. Byron wrote: "Though her years were waning, her climacteric teased her like her teens".

Hormonal changes

The menopause marks the pausing of menstruation, although hormonal alterations will have been taking place during the years prior to cessation, and may well continue for another two or more years. In a menstruating woman the menstrual controlling centre situated in the hypothalamus secretes releasing factors, which act on the pituitary gland resulting in the output of follicle-stimu-

lating hormone (FSH) which causes ovulation, and the output of luteinising hormone (LH) which stimulates the production of progesterone. These result in the ovarian production of oestrogen and progesterone, which in turn stimulates the endometrial lining of the uterus. There is a feedback mechanism from the ovaries and uterus informing the menstrual controlling centre of the state of the uterus. However, at the time of the menopause the ovaries stop ovulating and there is a deficiency of oestrogen and progesterone. The menstrual controlling centre reacts to this deficiency by increasing the stimuli and the pituitary increases the output of FSH and LH. This increased output, which can reach 10 times the level in a normal menstruating woman, remains raised for some 20 years until senility occurs. The menopausal symptoms are caused by the deficiency of ovarian hormones and the increase in FSH and LH.

However, these menopausal changes

do not occur abruptly, any more than the changes of puberty occurred overnight. They are gradual changes spread over several years, with the last menstruation marking the approximate half-way stage. The last menstruation may be heralded by a gradual shortening of menstrual loss, by decreasing menstrual flow, by missed menstruations, or by a combination of these alterations from the normal menstrual pattern. An abrupt ending, without previous decreased menstrual flow or missed menstruation, is rare and tends to occur if there has been a sudden change in a woman's life situation coinciding with the time of the hormonal changes; such events include moving house, bereavement, or following childbirth. It is this type of ending which is most likely to be accompanied by marked symptoms.

The mean age of the menopause in Britain is 49 years, with a range from 45 to 55 years. It tends to be earlier in those who had a late menarche, in nulliparous women, and those who experienced infertility. There may be a family pattern of early endings, as when mother and older sisters have all enjoyed an early change of life. Recent work in Boston has revealed an early menopause among heavy smokers.

Symptoms

The menopausal symptoms are of two types: the specific oestrogen deficiency ones which respond rapidly to oestrogen replacement, and the non-specific symptoms which may or may not be relieved by oestrogen (Table 1). The characteristic symptom, which few women avoid, is the hot flush, or "flash" as it is known in America. It is a sudden sensation of heat sweeping up from the waist to the face, accompanied by profuse sweating and blushing. If blushing is present the flushes become visible to bystanders, but some women experience invisible flushes. The sensation lasts only a few minutes and may be accompanied by panics, anxiety, or palpitations. At night, frequent waking in a bath of sweat may lead to insomnia and fatigue.

Thinning of the vaginal epithelium may lead to frequency of micturition (often misdiagnosed as cystitis), pruritis, dysuria, and dyspareunia with resultant loss of libido. The vagina appears pale and dry and the cervix firm and shrunken. The skin becomes thinner allowing wrinkles to appear.

Oestrogen deficiency may affect the bones, causing thinning or osteoporosis. This is initially manifested by fleeting pains, especially in the hands and feet with thickening of the interphalangeal joints. Leonardo da Vinci was the first to point out that the normal proportions

continued over

Table 1: Menopausal symptoms.

Specific	Non-specific
1. Hot flushes and night sweats.	Fatigue
2. Vaginal atrophy causing pruritis, urethral syndrome, dyspareunia, and loss of libido.	Insomnia
3. Thinning of the skin causing wrinkles.	Irritability
4. Pains in the bones, joints, and muscles, especially of the hands and feet.	Depression
	Headaches
	Palpitations
	Anxiety
	Vertigo
	Loss of memory and forgetfulness

are for the height to equal the armspan (Figure 1). However, after the menopause there may be a decrease in height without a decrease in armspan, and if the difference is marked oestrogen replacement is firmly indicated. At the later stage there is an increased susceptibility to fractures, especially of the wrist, neck of femur, and crushed vertebrae producing the characteristic widow's hump.

It is the nonspecific symptoms which many find most troublesome, particularly those who have previously boasted of their good health and freedom from neurotic traits. Until now they have had little patience with those complaining of fatigue, insomnia, irritability, and depression, headaches, anxiety, forgetfulness, and loss of memory.

The most annoying of Nature's changes associated with the menopausal years are those affecting the body contour, with the shrinking breasts and increasing weight tending to concentrate at the waist and abdomen, giving rise to the spare tyre and middle-age spread. It is at this time when weight swings predominate that the fat tend to get fatter and the thin become scraggy. It is now that full dividends are paid to those whose dietary habits succeed in controlling unwanted weight gain.

Sufferers of premenstrual tension are apt to find an increase in symptoms during the premenopausal years, with attacks of symptoms occurring even at the times of missed menstruation. But, as compensation, once the hormonal changes of the menopausal years are completed the familiar mood swings disappear and give way to a calmer and more relaxed personality. Those afflicted with menstrual migraine preceded by flashing lights, blurred vision, and other unpleasant visual aura often find that after the menopause only the aura remains, and gradually over the next few months the migraines themselves disappear.

While some women experience pain on coital penetration, due to the dryness and thinning of the vagina, the menopause does not mean the end of sexual activity; indeed, some find increased sexual pleasure with the end of contraception and the end of their premenstrual mood swings. Men do not have the abrupt hormonal changes experi-

enced by women. Their hormonal output shows a gradual decline between 20 and 70 years, and they continue to produce fertile sperm well past their three score years and ten. The age at which an individual male loses his sexual drive has unfortunately come to be called, quite wrongly, the male menopause. Loss of sexual drive can occur from any cause and at any age.

Women who have undergone a hysterectomy and/or oophorectomy will have an artificial menopause within days of the operation, although some are fortunate to be given an oestrogen implant at the time of the operation to ease their symptoms. The menstrual controlling centre will be confused by the sudden interference with the feedback pathway. In younger women, after about six to 12 months, the hot flushes will disappear and they may experience their usual premenstrual symptoms as the menstrual controlling centre reasserts its normal rhythm. Nevertheless these women will still have a natural menopause at the time they would normally have done so, and will again experience their hot flushes and psychological symptoms.

Hormone replacement therapy

Oestrogen is the hormone involved in hormone replacement therapy (HRT) at the menopause. It is amazingly simple and effective, with its benefits being appreciated within days of starting treatment. Few will deny the spectacular beneficial changes which occur in women with specific menopausal symptoms. The flushes disappear, sleep is restored, the vagina gradually becomes moist and brighter, and there is an accompanying general feeling of well-being and revival of vitality; gradually the non-specific symptoms vanish, or at least disappear into the background.

Oestradiol is the main oestrogen produced by the ovary during menstruating life, while oestrone, a much weaker oestrogen, is the one produced in the adrenals and peripheral tissues by postmenopausal women. Oestrogens used therapeutically include those with a steroid nucleus, such as oestradiol and oestrone, and the non-steroidal synthetic oestrogens such as stilboestrol, hexoestrol, and dienoestrol. This latter group is falling out of favour, especially since it has been appreciated that girls whose

mothers received the synthetic non-steroidal oestrogen, stilboestrol, during pregnancy are liable to develop adenocarcinoma during puberty. There is no shortage of natural oestrogens on the market with the pharmaceutical companies falling over themselves to satisfy an insatiable demand. The size of the potential market can be appreciated when it is realised that one-third of all women in Britain today are either menopausal or postmenopausal.

To avoid the possible risk of endometrial carcinoma developing after some 20 or more years of oestrogen therapy, it is advisable to use cyclical oestrogen therapy to allow intermittent shedding of the endometrium. The usual course of oestrogen lasts 21 days with seven days break during which breakthrough or very scanty bleeding may occur, but those who previously had longer menstrual cycles may prefer four weeks on tablets and one week off. If bleeding does not occur after three courses, a progestogen can be added either throughout the whole course or just for the last few days. Those who have had a hysterectomy and/or oophorectomy can have continuous oestrogen. Those contemplating long-term oestrogen replacement therapy should be seen at regular intervals for cervical smears, blood pressure and weight recording, breast palpitation, and any other investigations which may be indicated.

Contraindications

Oestrogen therapy is contraindicated in those with a history of coronary thrombosis, pulmonary embolism, or deep vein thrombosis, a hormone-dependent carcinoma of the uterus, ovary, or breast, diabetes, liver disease, or hypertension. Those with a family history of hypertension or hypercholesterolaemia should have a lipid study performed before embarking on HRT. Heavy smokers are also at risk and the choice may be left to them after the odds have been pointed out. Most patients in whom oestrogens are contraindicated can nevertheless respond to progestogens, such as norethisterone, and these stop the flushes and also prevent further deterioration of osteoporosis.

While few would deny oestrogen therapy to healthy women experiencing menopausal symptoms, the controversy surrounds the prophylactic use of oestrogens in those without symptoms and questions their prolonged use among symptom-free women in their 60s and 70s. Women with osteoporosis would be advised to continue therapy, but it needs to be appreciated that oestrogens will not correct any osteoporosis already present. The long-term effects of HRT will take many years yet to evaluate fully and until then no universal policy can be adopted.